

Deep brain stimulation for treatment of tremor in Parkinson disease

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University of Reykjavik 2020/2021

Contents

1. Introduction 2
2. A historical prospective 2
2.1 The first assumptions 2
2.2 The first electrical stimulator for pain treatment 3
2.3 Developing of deep brain stimulation (DBS) 3
3. Parkinson's disease 5
3.1 Brief description 5
3.2 Brain structures involved in PD 5
4. Deep brain stimulation 7
4.1 Stimulation parameters and voltage distribution in the Deep Brain Stimulation model 7
4.2 Overall architecture detection DBS 12
4.3 Implant manual: Lead Kit for Deep Brain Stimulation 3389, Medtronic 13
5. Conclusions 15
6. References 16

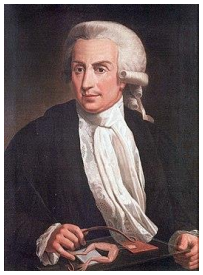
1. Introduction

The aim of this work is to explore the deep brain stimulation for treatment of tremor in Parkinson disease. I will go through the history of the first brain stimulation system reaching the first assumptions and applications.
Then I will cover Parkinson disease, the application field, and a commercial device used for this application and produced by Medtronic.
Thus, I will move on the electronic and instrumentation design analysing the keys of the progress on time and the main researchers who gave a significant contribution to this study.
As a master's student in biomedical engineering, I strongly believe that the role of my position and those of my colleagues, is to make possible what physicians suggest, creating a community that can move forward over time; in this work my community is the set of sources listed at the end.

2. A historical prospective

2.1 The first assumptions

Neuromodulation, the key word of this work, was first recorded for the treatment of pain occurred in about 15 AD, when Mr Scribonius, after observing that gout pain had been relieved by the accidental contact with a torpedo fish, recommended torpedo fish treatment for pain in general [1].



Picture 1 - Luigi Galvani

Investigations into this phenomenon were deepened in the 18th centuries when Galvani published an article (1971) appeared in the Proceedings of the Bologna Academy, reporting experimental results that, it was claimed, proved the existence of animal electricity. He found out that touching the muscle with a metallic object while the specimen lay on a metal plate provided the condition that resulted in the contractions [2], this incredible discovery, which happened by chance, was the first evidence of how electricity could influence the behaviour of muscles and nerves.
The first practical use of intraoperative electrical stimulation was performed by Sir Victor Horsley, who applied stimulation of tissue within an occipital encephalocele and noted conjugate eye movements in 1884. Two years later, he used stimulation to identify the thumb cortical area for local resection in a patient with focal epilepsy [3].



Picture 2 - Victor Horsley

2.2 The first electrical stimulator for pain treatment

Only at the beginning of the 20th century we find the first electrical stimulator, a home treatment machine called: ELECTREAT [4].

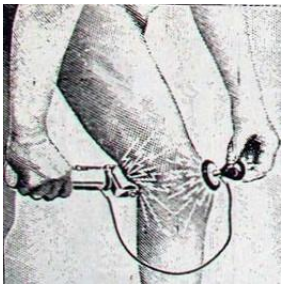
"Two things cannot occupy the same space at the same time.

Shooting electricity in forces the pain out."

From book cover of "ELECTREAT Mfg. Co. Peoria, Ill."

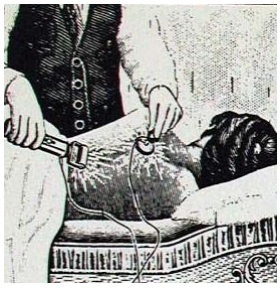
It was advertised to relieve not only pain, but every imaginable physical malady pain such as broken bones, bruises, boils toothache, and thus it had also a therapeutic function to get better aspects such as power for work, appetite, digestion, etc.
It was battery operated and had an uncanny resemblance to later transcutaneous electrical nerve stimulation (TENS) units.

The instrumentation is composed by a roller (positive pole) and a handle (negative pole), reaching the skin with both instruments, it closes the circuit, and it generates current.
As is well known, the current flows from positive to negative pole; the current direction, the way the circuit is closed, is extremely important because It changes the feeling and the treatment we are applying: the roller (+) constricts arteries and veins, relieves pain and reduce the inflammation while the handle (-) dilates arteries and veins, relaxes tissues and increase the inflammation.
The manual [4] shows all the body area which have to be reached in order to apply a specific treatment.



Picture 3 - Rheumatism

This instrumentation was a milestone for electrical stimulation, for the first-time people could relieve pain independently for many treatments.
This technology has developed until today by changing in electromagnetic machines that offer the same expectations as the previously described instrument, with an obviously greater effectiveness.
Anyway, the current stimulation is still widely used such as what we are analyzing in this study: electrical stimulation for treatment of tremor in Parkinson disease.



Picture 4 - Spinal treatment

2.3 Developing of deep brain stimulation (DBS)

In this paragraph I illustrate the main events and researchers that have characterized the development of DBS.

1963: Medtronic produced "Barostat", the first implantable pacemaker, which was used to stimulate the carotid sinus for the treatment of hypertension.



Picture 5 – first logo Medtronic (1949)

1965: The concept that provided the impetus for introducing neuromodulation for pain treatment was the introduction of Melzack and Wall's [6] landmark theory in 1965. They proposed that pain perception involves a gate that can be opened or closed depending on the balance between firing of small and large neural fibers. If large touch fibers were stimulated ("If you rub it, it feels better"), the gate would close and pain would lessen.
The application involved into electrical stimulation where the electrodes taped to the skin [7], this is so-called transcutaneous stimulation, later called TENS.
The first implantable stimulator used to treat chronic pain, was developed by Avery and Wepsic [8].



Picture 6 – first implantable pacemaker, Medtronic (1959)

…in the same year…

1965: Thomas Mortimer, a graduate engineering student, applied for a job interview at Medtronic to see if his cardiovascular stimulator might be adopted for dorsal column stimulation.
Medtronic produced "Angiostat", which stimulated the carotid sinus nerve for treatment of angina.

1967: Norman Shealy had the idea to stimulate the large nerve fibers where they were uniquely gathered in the dorsal columns of the spinal cord.
Norman Shealy recruited Thomas Mortimer to design an implantable stimulator. The first model required an external power supply connected by needles passed through the skin. Mortimer already had experience about this project cause he went to Medtronic to reach it.

1968: "Baristat" and "Angiostat" were medicated with platinum electrodes (originally made by Mortimer) obtaining the first spinal cord stimulators which were produced and commercialized by Medtronic in 1968. These stimulators came in two parts: the implantable portion consisted of an electrode connected to a circular wound antenna implanted subcutaneously while the external part had a battery-operated power supply and control box connected to a similar external antenna taped on the skin overlying the internal antenna.

1973: Shortly after the introduction of spinal cord stimulation, Hosobuchi [9] reported the successful use of chronic stimulation of the somatosensory thalamus for the treatment of denervation facial pain, anesthesia dolorosa, and the field of deep brain stimulation (DBS) was born.

1977: The spread of DBS led the Food and Drug Administration (FDA) to establish a symposium on its safety and efficacy. After a while, they felt that the use of DBS for motor disorders and pain management had not been sufficiently documented, DBS for pain management was de-approved, as it remains today, although the recent surge in DBS activity has brought renewed interest in that field.

Currently: DBS has been approved by the Food and Drug Administration as a treatment for essential tremor and Parkinson's disease since 1997 (FDA, Retrieved May 23, 2016) when The first device, Medtronic's Activa Deep Brain Stimulation Therapy System, was approved for tremor associated with essential tremor and Parkinson's disease.

3. Parkinson's disease

In this chapter, I will give a brief overview of the disease focusing then on the brain structures involved in the disease, useful for a better understanding of electrodes placement in deep brain stimulation.

3.1 Brief description

Parkinson's disease (PD) is the second most common neurodegenerative disease in the world; it affects an estimated 5 million people for year.
Parkinson's disease is a long-term degenerative disorder of the central nervous system that mainly affects the motor system with affects such as tremor, rigidity, slowness of movement, and difficulty with walking. Patients may also have problems with sleep and sensory systems, in addition, this disease has a strong psychological impact.
The motor symptoms of the disease come from the death of cells in the substantia nigra, a region of the midbrain, leading to a dopamine deficit.
The cause of death of dopaminergic neurons in the SNpc is unknown, however, those with a family member affected by PD are at an increased risk of getting the disease, other risk factors are those who have been exposed to certain pesticides and who have prior head injuries.
There is no cure for PD but just treatment aims to improve the symptoms; it is initially treated with inhibitory drugs or dopamine agonists, by time these medications become less effective and surgery to place microelectrodes for deep brain stimulation take place.

3.2 Brain structures involved in PD

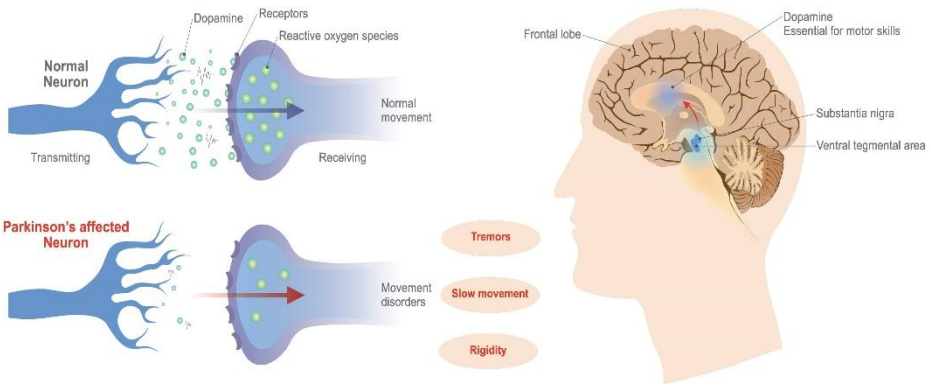
As mentioned above, PD comes from the death of cells in the substantia nigra (gif. 1) [13], a region of the midbrain.
The midbrain lies at the base of the brain and is the top part of the brain stem, it contains several collections of nerve cells called basal ganglia which help perform and relax voluntary muscle movements, suppress involuntary movements, and coordinate posture changes, including the substantia nigra.
The basal ganglia plays a key role, It normally exert a constant inhibitory influence on a wide range of motor systems, preventing them from becoming active at inappropriate times; dopamine acts to facilitate this release of inhibition, so high levels of dopamine function tend to promote motor activity, while low levels of dopamine function, such as occur in PD, demand greater exertions of effort for any given movement [12].



gif 1 - Substantia nigra highlighted in red

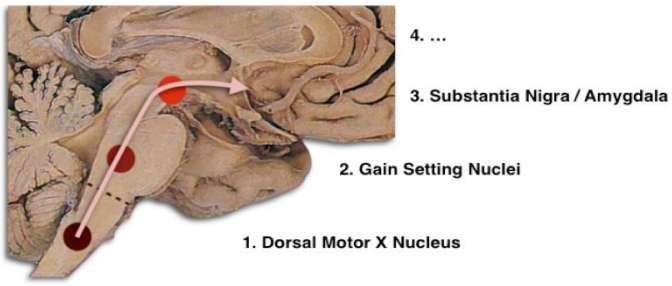
The substantia nigra is dark in color and contains nerve cells that release the neurotransmitter dopamine; the disease arises precisely from the deficit of this dopamine.

Picture 7 - dopamine path in a normal and Parkinson's affected neuron



There are five major pathways in the brain connecting other brain areas with the basal ganglia: motor, oculo-motor, associative, limbic and orbitofrontal circuits. All of them are influenced by Pd although the motor circuit is the most studied.

There are several mechanisms that can lead to the death of neural cells, one mechanism consists of an abnormal accumulation of proteins within the neurons (Lewy bodies); observing the position in time of bodies is possible to estimate the three main places of neuronal degeneration in PD which correspond temporally to the aggravation of the disease:



Picture 8 - progression of Lewy body deposits in PD

1. Olfactory bulb, medulla, oblongata and pontine tegmentum
2. Substantia nigra
3. Neo-cortex

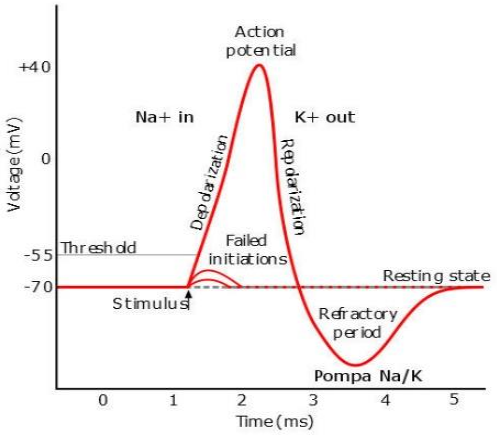
Knowing the brain areas responsible for the disease is fundamental for the application of DBS, in the next chapter I will deepen this field of application giving first an overview about its physiognomy and then moving on to the instrumentation used.

4. Deep brain stimulation

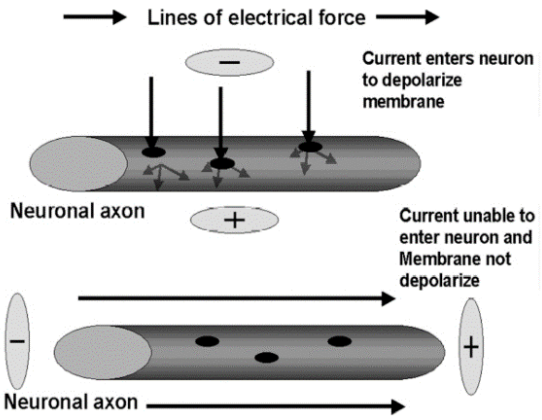
The motor symptoms of PD appear to result largely from abnormalities in one of the several parallels and largely segregated basal ganglia thalamocortical circuits (i.e., the motor circuits) in the human brain, the dysfunction of one or more of these circuits singly or in combination result in the disruption of downstream network activity in the thalamus, cortex, and brainstem [14].
In the following paragraphs I will go through the stimulations models and voltage distribution of the DBS, finally presenting a business model proposed by Medtronic.

4.1 Stimulation parameters and voltage distribution in the Deep Brain Stimulation model

The DBS has the role of reactivating these channels through subcutaneous electrical impulses so that they are able to function relatively more normally.
In details, neuronal activation requires generation of an electrical impulse across the cell membrane that is transmitted down the axon to the next neuron.
The potential difference between the inside and the outside of the membrane is influenced by the opening and closing of some protein channels, a dysfunction of these channels can develop the PD disease, in this case DBS acts to depolarize the cell membrane, to open the channels and create an electrical impulse: the more current will be driven into the neuron, and there will be greater depolarization and greater probability of generating an axon potential. It can be resumed that by controlling the voltage we can control the volume of tissue subjected to effective stimulation and by controlling the pulse width we can control how many and which neuronal elements within the volume of voltage will be stimulated.
To make current flow from one end of the neuron to the other it is necessary to impose an electric field perpendicular to the direction of the neuron, this depends mainly on the type of stimulation that can be monopolar or bipolar.



Picture 9 – Action potential Neuron



Picture 10 – orientation electric field inside the neuron

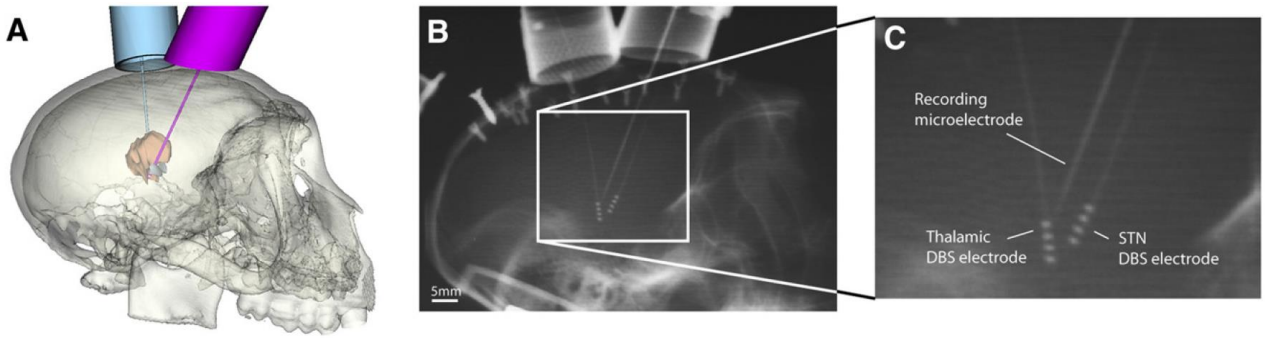
There are 3 basic parameters that must be taken into account when programming the DBS: 1) DBS electrode impedance, 2) capacitive modulation of the stimulus waveform and 3) inhomogeneity and anisotropy of the tissue medium. These factors influence the voltage which has been found responsible for understanding the electrical parameters of DBS and characterizing its effects on the nervous system [16].
Characterize the voltage distribution generated in the brain is the first step in predicting to the effects of DBS.

A study from the Department of Biomedical Engineering in Cleveland (USA, 2008) [16], reached the previous aspects related to the DBS by observing its effects in vitro, in vivo, and in the thalamus and subthalamic nucleus (STN) of a rhesus monkey, the goal was to record the voltage by placing a microelectrode around the DBS electrode up to a radius of 7 mm from it, then the results were compared to the one recorded with a volume conductor electric field model.



Picture 11 - rhesus monkey

Techniques of imaging such as Magnetic resonance imaging (MRI), Computed tomography (CT) and X-ray were used to plan the surgery, to estimate the electric tissue conductivity through the calculated density gradient and to check the reciprocal position between DBS electrode and microelectrode during the experiment. Thus, several software were used for surgical procedure and DBS electrode implantation such as Cicerone, a neurosurgical navigation software which can find the right coordinates for the installation of the electrode.

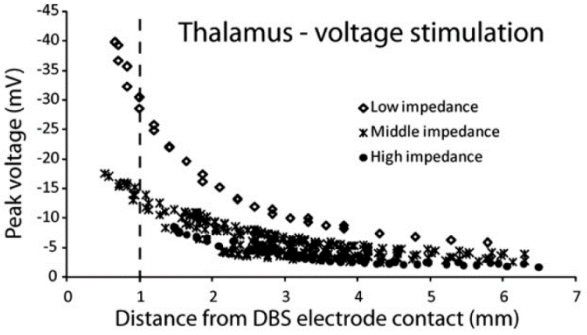


Picture 12 - (A) Cicerone software was used to plan locations for chambers targeting thalamus (left hemisphere) and STN (right hemisphere). (B) X-ray image taken during a recording session. (C) Zoomed in view of figure B.

This study shows that both in vitro and in vivo, voltage-controlled waveforms decayed exponentially but the vivo voltage decay occurred with a longer time constant than observed in vitro.

The maximal voltages were recorded near the active electrode contact and decayed non-linearly as a function of distance from the active electrode contact, it is due to the interface voltage drop which has been documented in the electrochemistry literature and is related to the transition from electrons carrying the current in the electrode to ions carrying the current in the medium; this result can be used to reduce the effective stimulation voltage applied to the model: for monkey DBS electrodes the drop was calculated to be 57% meaning that the 0.3 V stimulus created by the pulse generator resulted in an equivalent stimulus of 0.13 V after the interface voltage drop [16].

A change in DBS voltage was also observed, it changes as a function of the impedance of the DBS electrode, and It comes out mainly in the thalamus-voltage stimulation (see picture 13).
The results in vivo and in vitro were compared to the simulation model, interestingly, the more distant recording tracks in the STN showed greater errors, with the model overestimating the voltage, while the average error of the thalamic model was independent of distance from the active contact. It is probably due to conductivities of the STN model did create a more dramatic voltage gradient in that highly anisotropic region. This problem can be avoided by using current-controlled pulses which have stimulation thresholds that are effectively independent of electrode impedance.



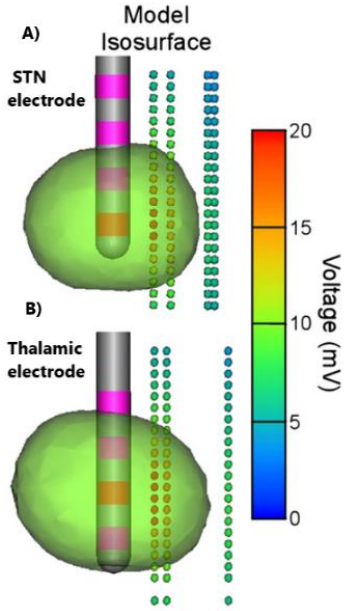
Picture 13 – DBS voltage in the thalamus; low DBS electrode impedance was defined as less than 3 kΩ, medium impedance as 3–5 kΩ, and high impedance as more than 5 kΩ.

A model voltage iso-surface around the DBS electrode has been calculated for both brain regions: It shows different region stimulated which is due to different conductivity regions.

We can resume that the DBS potential is related to the medium which is:

1. inhomogeneous, meaning that the extracellular space has different electrical conductivity
2. anisotropic, meaning it changes as a function of direction
3. directly affected by these different features of the electrode–brain interface.

This study provides the first experimental characterization of the voltage distribution generated by DBS electrode and substantiate the use of computational models to quantify the spread of stimulation as a function of the electrode placement and stimulation parameter settings used in an individual subject.

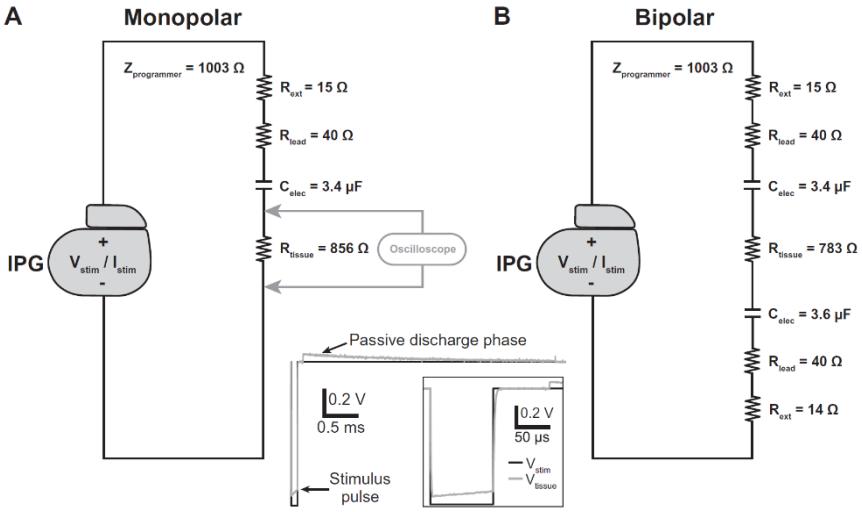


Picture 14 – Voltage around STN and Thalamic DBS electrodes

Another study from the same Department of Biomedical Engineering in Cleveland [18], defined an equivalent circuit model that accurately captured the output of the IPG. They compared the detailed circuit model of the clinical stimulus waveforms to simplified representations commonly used in computational models of DBS.
The purpose of this study was to determine a circuit model for DBS stimulation by calculating all setting parameters one at a time holding the other parameters constant (see tab.1).

	Voltage amplitude (V)	Current amplitude (mA)	Pulse width (μs)	Stimulation configuration	Pulse frequency (Hz)	Electrode impedance (KΩ)
Pulse amplitude	1 2 4 6 8 10	1 2 4 6 8 10	90	monopolar	130	1
Pulse width	1	1	90 450	monopolar	130	1
Stimulation configuration	1	1	90	monopolar bipolar	130	1
Pulse frequency	1	1	90	monopolar	30 130 250	1
Electrode impedance	1	1	90	monopolar	130	0.5 1 1.5

Tab. 1 – programming parameters of DBS stimulator

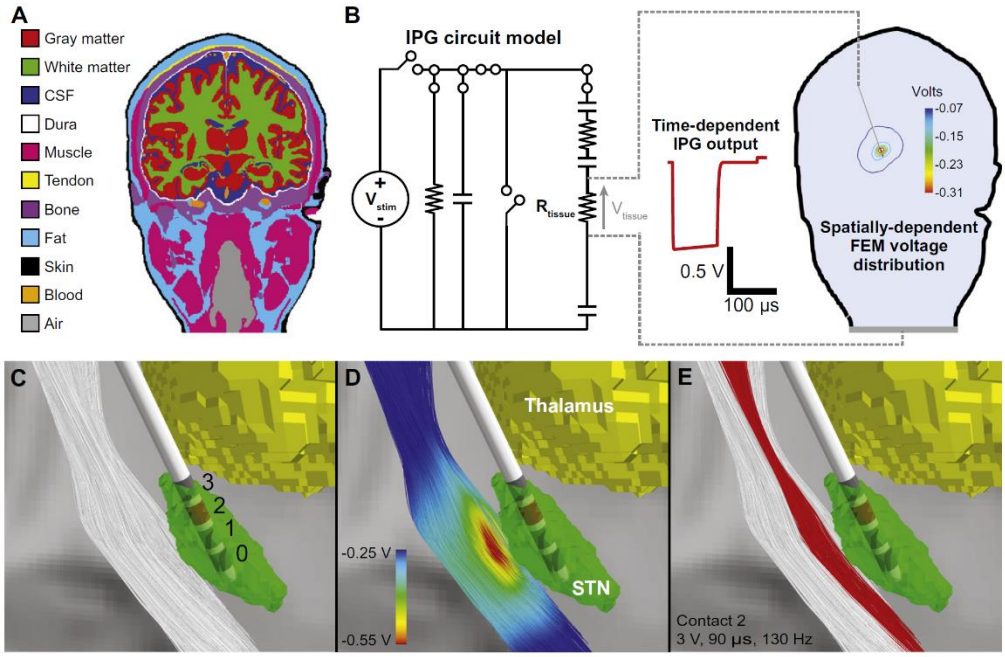


Picture 15 – A (monopolar) and B(bipolar) IPG model

The implantable pulse generator (IPG model) includes: C_{block}(blocking capacitors at the IPG outputs), $R_{ext+lead}$ (resistance of the wire which connects the lead), R_{tissue}, C_{elec}(electrode-tissue interface capacitance), R_p(parasitic resistance) and C_p(parasitic capacitance).

These circuits were used to measure the IPG output for both voltage-controlled and current-controlled stimulation. The waveform at the bottom (picture 15) shows an example bench recording of V_{tissue} for monopolar voltage-controlled stimulation for pulse amplitude (i.e. V_{stim}) = 1 V, pulse width = 90 us, and pulse frequency = 130 Hz.

To estimate the tissue voltages generated by DBS, they scaled the electrostatic 3D spatial voltage distribution (from the FEM) by the time-dependent V_{tissue} (from the IPG circuit model). Then, they applied these time- and space-dependent tissue voltages to the axon models to determine the neural response to DBS. Thus, they used the extracellular mechanism within NEURON, to apply the scaled voltage distributions to each axon.

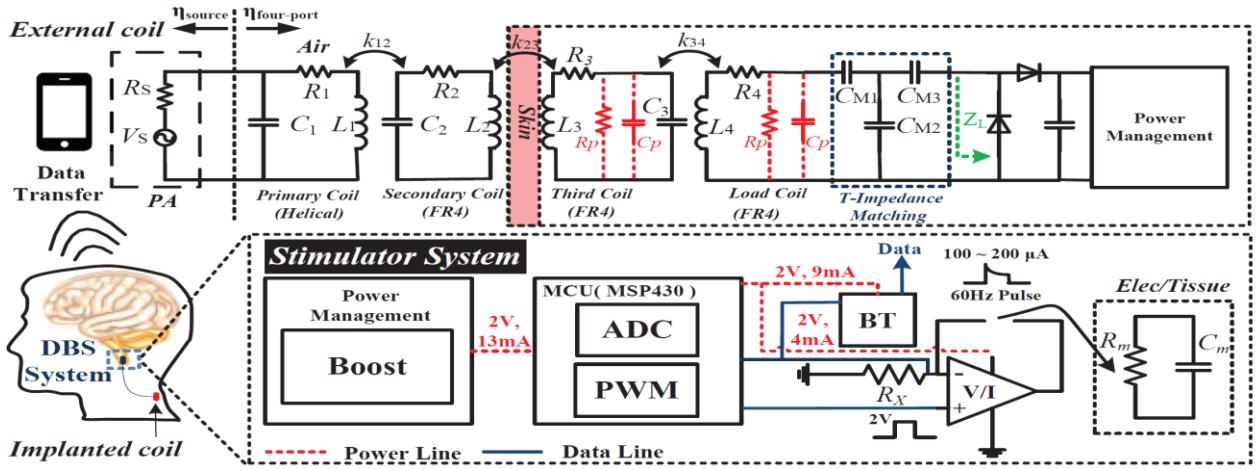


Picture 16 – FEM model of DBS and its voltage distribution

Finally, we see (A) a 3-dimensional finite element model (FEM) of DBS using an anatomical model of a human head and neck, (B) Spatially-dependent FEM voltage distribution and (C,D,E) show how the electrode interacts with the fibers of the nervous system.

4.2 Overall architecture detection DBS

The following closed loop permits detection and treatment of the neuronal cells of the patient. The disease symptoms can be improved at a microstimulator intensity of approximately 100‑200 μ A and the implantable integrated chip can be embedded beneath the skin to a depth of 5 to 15 mm.
Let's go over the main building blocks of a possible configuration of this stimulator [17].



Picture 17- closed-loop DBS stimulation system

T-match impedance matching circuit is one of the circuits used to match the impedance between two points, in this case, a source and a load. It increases the maximum **wireless power transfer** delivery to the load decreasing the reflection coefficient.
The front end of the power management receives ac power from the four coils, and then converts ac power into dc voltage through the voltage doubler, which can provide a V_{DD} of 2 V and I_{max} = 1.5 mA for the microprocessor control unit (MCU).
The stimulation system converts the tension in pulse current reaching a stimulation intensity of approximately 100–200 µA, meanwhile it samples the RX voltage and Convert it to the equivalent current, which became the incoming voltage to the ADC. Finally, the data were transmitted via Bluetooth for real-time monitoring on a phone. The stimulation frequency ranged 10 ‑ 100 Hz, at a pulse-current intensity below 200 μ A, and the electrical stimulation period lasted approximately 5 ‑ 500 μ s for the treatment.

The main stimulation parameters are active contacts, voltage, pulse width, and frequency, they are set according to the patient and how the disease manifests itself.
It comes out that DBS programming is important for cost-effective and timely care, the guide DBS management is based on electrophysiological properties and the regional functional anatomy surrounding the DBS lead.

4.3 Implant manual: Lead Kit for Deep Brain Stimulation 3389, Medtronic

In this paragraph I will deal with a commercial DBS device going through its implant manual highlighting all the parts that I think are most relevant to understand the mechanics of its implantation and use.
"The Medtronic Activa System 3389" [19] is an implantable, multiprogrammable system that delivers electrical stimulation to selected areas of the brain, it can be used for several diseases such as Parkinson. It plays a bilateral stimulation of the internal globus pallidus (GPi) or the subthalamic nucleus (STN) in order to reduce some of the symptoms of advanced, levodopa-responsive Parkinson's disease that are not adequately controlled with medication. Thus, the system is intended for use in patients who are diagnosed with Essential Tremor or Parkinsonian tremor not adequately controlled by medications and where the tremor constitutes a significant functional disability.

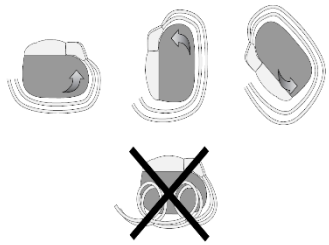
About contraindications: Implantation of an Activa Brain Stimulation System is contraindicated for patients exposed to diathermy where energy from diathermy can be transferred through the implanted system and can cause tissue damage at the location of the implanted electrodes, resulting in severe injury or death.
It is not allowed to the patients who will be exposed to Magnetic Resonance Imaging (MRI) using a full body transmit radio-frequency (RF) coil, a receive-only head coil, or a head transmit coil that extends over the chest area. Performing MRI with this equipment can cause tissue lesions from component heating, especially at the lead electrodes, resulting in serious and permanent injury including coma, paralysis or death. Patients for whom test stimulation is unsuccessful and/or are unable to properly operate the brain stimulator are also not allowed to receive this treatment.

About the programming parameters: suppression of symptoms should occur at amplitudes of 1 to 3.5 V, pulse widths of 60 to 120 µsec, and rates of 130 to 185 Hz. Higher amplitudes and pulse widths may indicate a system problem or less than optimal lead placement.
Lead DBS store temperature should be between -30° F (-34° C) and 135° F (57° C). Temperatures outside this range can damage components.
Stimulation settings for systems implanted in the internal Globus Pallidus (GPi) may be higher than stimulation settings for systems implanted in the Subthalmic Nucleus (STN), consequently, systems implanted in the GPi may have shorter battery life than systems implanted in the STN, thus, the long-term safety and effectiveness of Activa Therapy has not been established.
If more than one neurostimulator is implanted, then the potential for unintentional programming changes to the other neurostimulator exists. If two neurostimulators are implanted, they must be implanted at least 8 inches apart to minimize interference. Verify final programmed parameters by reviewing both devices at the conclusion of any programming session.

Programmable Parameters[a,b]	**Values**
Pulse Amplitude	Normal Resolution: 0 to 10.5 V Fine Resolution: 0 to 6.35 V
Rate	66 values from 3 to 250
Pulse Width	60, 90, 120, 150, 180, 210, 240, 270, 300, 330, 360, 390, 420, 450 µsec

Table 2 - Programmable Stimulation Parameter Values

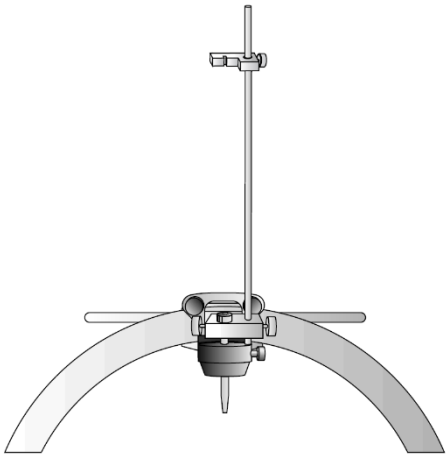
About the warnings: It is mandatory wipe off any body fluids on the extension or lead contacts or connector before connecting.
Contamination of connections can cause intermittent stimulation or shorts in the neurostimulation circuit. Thus, limit counter-clockwise rotations of neurostimulator setscrews (see picture 18), too many counter-clockwise rotations may disengage the setscrew from the connector block. Place the neurostimulator away from bony structures, this also helps to minimize the possibility of skeletal muscle stimulation, which may be perceived by the patient as twitching or burning.
Wrap any excess extension wire around the perimeter, this avoids any increase in subcutaneous pocket depth, helps minimize potential damage during neurostimulator replacement surgery, and helps minimize potential kinking of the extension wire. Handle the implanted components of this system with extreme care, these components may be nicked, cut, or damaged by excessive traction or sharp instruments and may require surgical replacement.



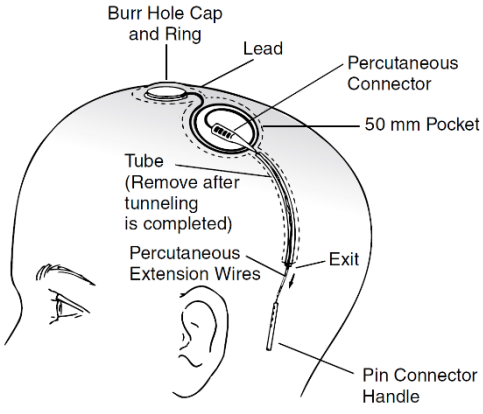
Picture 18 – warnings about the lead position

About Electromagnetic Interference (EMI): medical and environmental (home, occupational, and other) devices may generate electromagnetic interference which change the parameters of a neurostimulator; turn a neurostimulator off and on; or cause a neurostimulator to surge, shock, or jolt the patient.
The Activa System may affect the operation of other implanted devices, such as cardiac pacemakers and implantable defibrillators, in this case careful programming of each system may be necessary to optimize the patient's benefit from each device.
Electrocautery can damage the lead, thus use of bipolar electrocautery is recommended.
Other sources of damage might be External Defibrillators, High Radiation Sources, Lithotripsy, Psychotherapeutic Procedures and many other applications.

About implantation: DBS Lead requires stereotactic neurosurgical techniques for the initial implant and ongoing postoperative patient management, the target site may be localized for stereotactic implantation of the DBS Lead using CT scans, MRI, or ventriculography. A test stimulation or mapping electrode may then be utilized for further localization of the target with electrical stimulation. To implant the DBS Lead, an insertion cannula and stylet should be placed to a point approximately 15 mm proximal to the target site for stimulation. The lead should be passed through the insertion cannula and advanced to the target site.

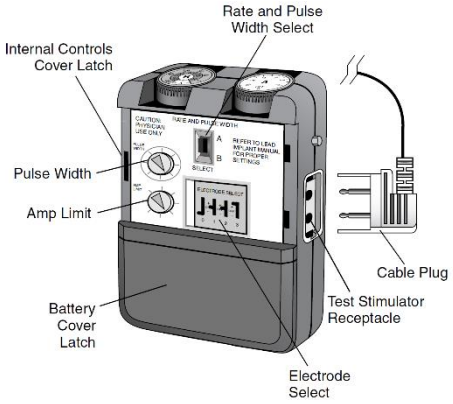


Picture 19 - Attach lead holder assembly to frame

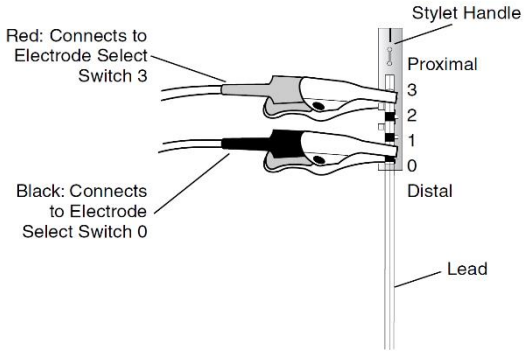


Picture 20 - Tunneling and lead placement

After the implementation: the electrode is tested several times with a test simulator, attaching the alligator clips to the applicable lead contacts correspond to the desired electrodes. The test is carried out both to verify the correct functioning of the lead and to simulate the programming by intervening in case something goes wrong.
The model 3625 test stimulator allows to check the connection of the electrode with the target by gradually changing all the setting parameters such as pulse width, frequency, electrode select and amplitude voltage/current.



Picture 21 - Model 3625 Test Stimulator Internal Controls



Picture 22 - Connect clips to lead contacts

5. Conclusions

Deep brain stimulation is nowadays one of the most widely used methods for the alleviation of Parkinson's symptoms when they cannot be alleviated by classic medications. It is an invasive method that was approved only in recent years by FDA (food and drug administration).

In this chapter I reviewed the most important events that led to the development of this technology, then I moved on to the analysis of the Parkinson disease, and then focus on electronic instrumentation that is the field closest to my course of study.
In particular, I focused on the stimulation and voltage distribution mechanism which is the key for a good DBS programming.
From the articles that I have developed, it turns out that an analysis of the potential distribution is essential to attribute the levels of neural stimulus and then regulate the parameters of stimulation; optimize the choice of stimulation parameters, implies a more selective treatment on the highlighted symptoms and decreases the testing time during the intervention, as well as improving battery life by reducing the pulses only when and where it is necessary.

I also want to emphasize that this chapter takes a strictly engineering approach where we leave it to our medical colleagues to investigate the nature of the disease and all its controversies.

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